Advanced Topics in Monetary Economics with Heterogenous Agents

— Problem Set 3, Spring 2021 —

Due: Monday, March 1, 8:29 AM ET (before Section) on Canvas

Important guidelines.

1. Please only submit this problem set if you are *not auditing* this course and are *officially enrolled* for credit. Each problem set counts for 20% of your final grade.
2. Make sure to *submit your code*. The best way to solve our problem sets is using Jupyter Notebooks. We *strongly suggest using Python* from the beginning since this course will progressively build on existing Python modules in future problem sets. You can ask your classmates coding questions in Slack, Martin in OH, and review QuantEcon's programming tutorial.[1]
3. The problem sets for this class are meant to give you exposure to the actual, messy research process. This means: If a question is incomplete or missing information, please go ahead and make an assumption that you deem reasonable. Be sure to *state your assumptions and submit your full derivations* in your solution. Many problems could have multiple solutions.

1 The zero liquidity limit

In this problem, we explore the properties of the zero liquidity model a little more. We first think about how to calibrate a two-state model to hit a certain MPC. We then look for an analytical expression for the iMPCs and fiscal multipliers, and verify these on the computer.

1.1 Calibrating the model with two states

A very important calibration target for our models is the aggregate income-weighted MPC M. For the purpose of this exercise, let's say we try to hit an M of 0.546 (this is the value from the Norwegian data in the IKC paper). We also want to normalize aggregate income to $Y = 1$.

With two states, the Markov transition matrix is

$$\Pi = \begin{bmatrix} 1 - p_L & p_L \\ p_H & 1 - p_H \end{bmatrix}$$

[1] https://python-programming.quantecon.org/intro.html

and the values taken by the idiosyncratic productivity state (possibly after filtering through the progressive tax system) are s_L and s_H. This gives us four degrees of freedom: s_L, s_H, and the exit probabilities p_L, p_H.

1. Express the stationary distribution of the Markov chain for income (π_L, π_H) as a function of p_L and p_H

2. Calculate the β needed to sustain the zero liquidity limit as a function of p_H, s_L and s_H. Let $\overline{\rho} = \frac{1}{\beta(1+r)}$. *(Note: At this level of β, the Euler equation of the H agent holds with equality at the at the zero-liquidity allocation)*

3. By linearizing the Euler equation of the H agent, one can derive the following equation for the MPC m of the H agent

$$\frac{m}{1-m} = \frac{1+r}{\overline{\rho}} \left(\sum_{s'} \Pi_{\overline{s}s'} \left(\frac{s'}{\overline{s}} \right)^{-(\sigma+1)} + \Pi_{\overline{s}\overline{s}} m \right) \tag{1}$$

   Here $\overline{s} = s_H$. Using your results from 2, find an expression for the MPC of the H agent m as a function of only p_H, s_L and s_H

4. Describe how you would calibrate this model for given π_L and $\frac{s_H}{s_L}$ to hit our target for the aggregate income-weighted MPC $M = 0.546$ and $Y = 1$.

1.2 iMPCs for the zero liquidity model

We now want to derive an expression for the iMPCs of the zero liquidity model, $\frac{\partial C_t}{\partial Z_s}$, as defined in class. Of course, we want $\frac{\partial C_0}{\partial Z_0} = M$ as calibrated above, but we now ask how the other degrees of freedom of our calibration π_L and s_H/s_L affect the shape of iMPCs.

5. Considering the Euler equation of the agent at the top together with the budget constraints,

$$c_t\left(\overline{s}\right)^{-\sigma} = \beta\left(1+r\right)\sum \Pi_{\overline{s}s'} c_{t+1}\left(s'\right)^{-\sigma}$$
$$c_t\left(s\right) + a_t\left(s\right) = \left(1+r\right)a_{t-1} + sZ_t$$

   derive a second-order difference equation for the assets $a_t\left(\overline{s}\right)$ of agents in the top state $\overline{s}$. Aggregate into a difference equation for total assets A_t.

6. Solve this difference equation to obtain the evolution of aggregate assets. Hint: you should find that that there is a certain $\mu \in (0,1)$ and constant $\alpha_0 > 0$ such that:

$$dA_t = (1-m)(1+r)dA_{t-1} + (1-m)(1-\mu)dZ_t - \alpha_0 \sum_{u=1}^{\infty} \left(\beta(1-m)(1+r)\right)^u dZ_{t+u}$$

$$dC_t = m(1+r)dA_{t-1} + (\mu + m(1-\mu))dZ_t + \alpha_0 \sum_{u=1}^{\infty} \left(\beta(1-m)(1+r)\right)^u dZ_{t+u}$$

What is the implied "fake news" matrix $\mathcal{F}^{C,Z}$ of aggregate consumption with respect to aggregate income? How do our two degrees of freedom π_L and $\frac{s_H}{s_L}$ affect μ and α?

7. Verify your results numerically using the class notebooks.

1.3 Fiscal multipliers in the ZL model

8. Next we consider fiscal multipliers. Starting from your second-order difference in question 5 (5), derive an expression for dY_t as a function of dG_t, dT_t and the path of future debt. In particular, show that there exist constants $\alpha_1 > 0$, $\alpha_2 > \alpha_1$ such that

$$dY_t = \frac{1}{1-\mu}\left[dG_t - \mu dT_t\right] + \frac{1}{1-\mu}\alpha_1 dB_t + \frac{1}{1-\mu}\alpha_2 \sum_{k=1}^{\infty} dB_{t+k} \tag{2}$$

9. Verify your results numerically using the class notebooks.

Side remark. Equation (2) explains why we found a "fiscal policy forward guidance puzzle" in the zero-liquidity model: deficit financed policy far into the future has non-discounted effects in the present.

2 Fake news algorithm for firm dynamics

This question asks you to implement the fake news algorithm for the firm dynamics problem *without entry and exit* following up on the first part of questionn 2 in Problem Set 2.[2] Here is the streamlined version of the problem (refer to Problem Set 2 for the full description).

$$\max V_t\left(s, n_-\right) = \max_n \left\{ d_t + \frac{1}{1+r_t}\mathbb{E}\left[V_{t+1}\left(s', n\right) \mid s\right]\right\}$$

$$d_t = f\left(s, n_-\right) - w_t n_- - \Psi\left(n, n_-\right), \quad f\left(s, n_-\right) = X_t s^\alpha n_-^{1-\alpha}, \quad \Psi\left(n, n_-\right) = \frac{\psi}{2}\left(\frac{n}{n_-} - 1\right)^2 n_-$$

Aggregates dividends, production, labor demad, and investment costs, are all defined in the same way: weighted averages using the endogenous distribution of firms.

$$\Phi_t = \int d\left(s, n_-\right) dD_t\left(s, n_-\right) = Y_t - w_t N_t^f - \Psi_t$$

Set T to be equal to 100. Our goal is to compute the Jacobians of aggregate labor demand and output with respect to real wages, i.e.

$$\mathcal{J}^{N^f,w} \qquad \text{and} \qquad \mathcal{J}^{Y,w}$$

These Jacobians, much like the iMPC matrix $\mathbf{M}$ for the household problem, are the key ingredients to compute aggregate dynamics of the model to a first order.

[2] Reach out to maragoneses@g.harvard.edu if your code for that part is not working correctly.

2.1 Steady state

1. To begin, make sure you have computed the steady state by iterating backwards the steady state Bellman equation

$$\max\ V(s, n_-) = \max_n \left\{ d + \frac{1}{1+r} \mathbb{E}\left[V(s', n) | s \right] \right\}$$

using the endogenous gridpoints method, as on the previous problem set, to find the steady state employment policy

$$n^*(s, n_-)$$

From now on, we denote by $\mathcal{T}_w n$ the backward iteration with assumed wage w. That is, in the steady state, we have that

$$\mathcal{T}_w n^*(s, n_-) \approx n^*(s, n_-)$$

for the equilibrium steady state wage w you found in the last problem set.

Let $D(s, n_-)$ denote the distribution. We denote by $\mathcal{D}_n D$ the forward iteration that maps distribution D into next period's distribution. In the steady state then,

$$\mathcal{D}_{n^*} D^*(s, n_-) \approx D^*(s, n_-)$$

We denote aggregate steady state labor demand by N^{f*} and aggregate production by Y^*.

Another way to write the forward iteration, by using the grid representation is

$$\Lambda' \mathbf{D}^* \approx \mathbf{D}^*$$

where $\mathbf{D}^*$ is a vector storing the pdf of the distribution across the (s, n_-) idiosyncratic state space. Λ' is the steady state transition matrix. (We add a transpose here to Λ as transition matrices are typically row-stochastic.) Let $\mathbf{n}_{grid}$ be the vector that just as the value of the n_- grid at every state (s, n_-).

2.2 Direct algorithm

We begin computing the Jacobian $\mathcal{J}^{N^f, w}$ using the direct algorithm.

1. Iterate through the following steps, for each column ℓ from 0 to $T-1$, now feeding in a wage change at date ℓ (this represents the time horizon, which sometimes we also refer to s).

   i. Set the policy at the end date $n_T(s, n_-) = n^*(s, n_-)$.

   ii. Go backwards in time, computing

      i. for each $t \neq \ell$,

$$n_t(s, n_-) \equiv \mathcal{T}_w n_{t+1}(s, n_-)$$

ii. and when $t = \ell$

$$n_\ell(s, n_-) \equiv \mathcal{T}_{w+h} n_{\ell+1}(s, n_-)$$

where we choose some small h, e.g. $h = 10^{-4}$

iii. Using the policies, roll the distribution forward, computing D_t, $t = 0, 1, \ldots, T-1$, according to

$$D_{t+1}(s, n_-) = \mathcal{D}_{n_t} D_t(s, n_-)$$

where $D_0 = D^*$.

iv. Compute aggregates

$$N_t^f \equiv \int n_- dD_t(s, n_-)$$

v. Compute vector

$$\mathbf{v}_\ell \equiv \frac{N_t^f - N^{f*}}{h}$$

2. Plot the vector for columns $\ell = 0, 5, 10, 15$.

3. In practice, we can improve accuracy by subtracting a "ghost run". Follow the steps in subquestion 1 above, but for a single column setting $h = 0$ (i.e. only use steady state policies). Denote the resulting vector by $\mathbf{v}^*$. Set

$$\mathbf{v}_\ell \equiv \frac{N_t^f - N^{f*}}{h} - \mathbf{v}^*$$

Plot the vector again for columns $\ell = 0, 5, 10, 15$.

4. Time your algorithm.

2.3 Fake news method

We now use a more efficient algorithm.

1. Set the policy at the end date $n_T(s, n_-) = n^*(s, n_-)$.

i. First, do a ghost run. Compute the policies just like before, but without feeding in a shock, arriving at $n_t^*(s, n_-)$. Denote $D_{1\ell}^* \equiv \mathcal{D}_{n_{T-1-\ell}^*} D^*$ as the distribution we arrive at after one forward iteration for each one of the n_t^*'s.

ii. Now set $\ell = T-1$ and compute the policies just like before with a shock at $\ell = T-1$, arriving at $n_t(s, n_-)$. n_t corresponds to the policy when the wage changes in $T-1-t$ periods. This is just a single backward iteration.

iii. The first row of the Jacobian $\mathcal{J}^{N^f, w}$ is zero here. Why?

iv. Compute the change in the distribution after the first row, for any $s = 0, \ldots, T-1$ compute:

$$dD_s \equiv \mathcal{D}_{n_{T-1-s}} D^* - D_{1s}^*$$

2. Compute the *expectations vectors* as follows

$$\mathcal{E}_0 = \mathbf{n}_{grid}$$

$$\mathcal{E}_{t+1} \equiv \Lambda \mathcal{E}_t$$

$\mathcal{E}_t$ as the simple interpretation as the average level of employment a firm in a given state (s, n_-) today expects to have t periods from now.

3. Put together the fake news matrix

$$\mathcal{F} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ \mathcal{E}_0' dD_0 & \mathcal{E}_0' dD_1 & \mathcal{E}_0' dD_2 & & \mathcal{E}_0' dD_{T-1} \\ \mathcal{E}_1' dD_0 & \mathcal{E}_1' dD_1 & \mathcal{E}_1' dD_2 & & \mathcal{E}_1' dD_{T-1} \\ \vdots & \vdots & \vdots & & \vdots \\ \mathcal{E}_{T-2}' dD_0 & \mathcal{E}_{T-2}' dD_1 & \mathcal{E}_{T-2}' dD_2 & \cdots & \mathcal{E}_{T-2}' dD_{T-1} \end{bmatrix}$$

4. Compute the Jacobian iteratively, as

$$\mathcal{J}_{t,s} \equiv \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s}$$

5. Plot columns $s = 0, 5, 10, 15$. Compare to the direct approach.

6. Time the procedure.

3 DAG practice

This is still about the firm dynamics model, but does not require a solution to the previous question.

1. Imagine you would like to simulate the impulse response of the firm dynamics model to a TFP shock, i.e., a shock to X_t. Find a "DAG" that you can use to simulate this model, with two unknowns and a single exogenous variable (namely X_t).

2. Now instead, imagine you assume that nominal wages are rigid (just like in the canonical HANK model in class), and monetary policy sets an exogenous real interest rate path r_t. Find a DAG that suits this purpose, with a single unknown and a single exogenous variable (namely r_t).